

GENERAL AN ELEMENTARY DERIVATION OF THE HITTING TIME DISTRIBUTION OF BROWNIAN MOTION WITH DRIFT

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Summary: Let B_s , $s \geq 0$ be a standard Brownian motion. For given constants μ and $b \neq 0$, denote by T the time at which the motion $B_s + \mu s$, $s \geq 0$ first hits the barrier b . The derivation of the Laplace transform of T is standard fare in courses on continuous time stochastic processes. Typically, the method is to find first the Laplace transform of T when $\mu = 0$ and then to deduce the general result from this special case. In this note an *elementary* proof is given of the general result. A well-known explicit expression for the distribution function of T is found by using just one standard Laplace transform and some elementary calculus. The proofs used here could be useful in introductory post-graduate courses on stochastic processes and continuous time finance.

1. Introduction

Let B_s , $s \geq 0$ be a standard Brownian motion. For given constants μ and $b \neq 0$, denote by T the time at which the motion $B_s + \mu s$, $s \geq 0$ first hits the barrier b . That is,

$$T = \inf \{s \geq 0 \mid B_s + \mu s = b\}.$$

In the modern day theory of continuous time finance, the distribution of T arises, for instance, in the study of certain path-dependent options. The derivation of the probability distribution of T is standard fare in most textbooks

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on continuous time stochastic processes. Typically, the method is to find first the Laplace transform of T when $\mu = 0$ using strong Markov or martingale methods - see Breiman (1968, page 289) for an instance of the Markov method and Revuz and Yor (1991, Proposition 3.3.7) for the martingale method. Then, advanced methods of one kind or another are employed to deduce a result in the case where $\mu \neq 0$ - see Karatzas and Shreve (1991, page 196 together with Problem 5.8 and Exercise 5.10 on page 197) or Revuz and Yor (1991, Exercise 3.3.14). In section 2 below an *elementary* proof is given of the relation

$$E[\exp(-\alpha T) \mathcal{I}(T < \infty)] = \exp\left(\mu b - (\mu^2 + 2\alpha)^{1/2} |b|\right), \quad \alpha \geq 0, \quad (1)$$

$\mathcal{I}(\cdot)$ denoting the indicator function. The proof requires familiarity with the monotone convergence and bounded convergence theorems of probability theory but does not involve any advanced concepts from the theory of discrete or continuous time stochastic processes.

In Section 3 it is shown, using one standard Laplace transform and some elementary calculus, how (1) can be inverted to yield the distribution function

$$P(T \leq x) = \exp(2b\mu) \Phi(-|b|x^{-1/2} - \operatorname{sgn}(b\mu)|\mu|x^{1/2}) + \Phi(-|b|x^{-1/2} + \operatorname{sgn}(b\mu)|\mu|x^{1/2}), \quad x > 0 \quad (2)$$

where $\operatorname{sgn}(a) = 1, 0$ or -1 for $a > 0, a = 0$ and $a < 0$ respectively and Φ is the standard normal CDF. Finally, in section 4, it is shown how a number of results that are generally derived in a rather fragmentary manner in available textbooks follow directly from (1) and (2).

The present paper is of an entirely pedagogic nature. No new results are established. However, the proofs given here use elementary probabilistic concepts only and should be accessible to the typical Honours student in

Actuarial Science, Mathematical Finance or Statistics in a South African university.

2. Proof of equation (1)

Let $t > 0$ and γ be fixed constants. Set

$$Z_s = \exp(\gamma B_s - \gamma^2 s/2), \quad s \geq 0$$

and $\tau(t) = \min\{T, t\}$. Using the moment generating function of B_s , one sees that

$$E[Z_s] \equiv 1 \text{ for all } s \geq 0. \quad (3)$$

The derivation of (1) that follows rests upon the fact, demonstrated via Lemma 1 below, that the fixed s in (3) can legitimately be replaced by the random $\tau(t)$, that is,

$$E[Z_{\tau(t)}] = 1. \quad (4)$$

Observe that the event $(\tau(t) \leq t)$ is the union of the mutually exclusive events $(T \leq t)$ and $(T > t)$. Thus, from (4),

$$\begin{aligned} 1 &= E[Z_{\tau(t)}] \\ &= E[\exp(\gamma(b - \mu T) - \gamma^2 T/2) \mathcal{I}(T \leq t)] \\ &\quad + E[Z_t \mathcal{I}(T > t)] \\ &= \exp(\gamma b) E[\exp(-(\gamma \mu + \gamma^2/2)T) \mathcal{I}(T \leq t)] \\ &\quad + E[Z_t \mathcal{I}(T > t)]. \end{aligned} \quad (5)$$

We now focus on the last equality in (5). Given $\alpha > 0$, the equation

$$\gamma \mu + \gamma^2/2 = \alpha \quad (6)$$

has two solutions in γ , namely

$$\gamma_1 = -\mu + (\mu^2 + 2\alpha)^{1/2} > 0$$

and

$$\gamma_2 = -\mu - (\mu^2 + 2\alpha)^{1/2} < 0.$$

Hence, (5) can be recast in the form

$$\begin{aligned} & E[\exp(-\alpha T) \mathcal{I}(T \leq t)] \\ &= \exp(-\gamma b) - E[\exp(\gamma(B_t - b) - \gamma^2 t/2) \mathcal{I}(T > t)], \end{aligned} \quad (7)$$

$\gamma \in \{\gamma_1, \gamma_2\}$. Notice that, on the left-hand side of (7),

$$E[\exp(-\alpha T) \mathcal{I}(T \leq t)] \rightarrow E[\exp(-\alpha T) \mathcal{I}(T < \infty)] \quad (8)$$

as $t \rightarrow \infty$, by the monotone convergence theorem.

Now suppose that b is positive. Then the event $(T > t)$ contains the event $(B_t - b \leq -\mu t)$. Thus, taking $\gamma = \gamma_1$ (which is positive) we have from (6) that

$$\gamma_1(B_t - b) - \gamma_1^2 t/2 \leq -\gamma_1 \mu t - \gamma_1^2 t/2 = -\alpha t \quad (9)$$

on the event $(T > t)$, so that the expectation on the right-hand side of (7) lies between 0 and $\exp(-\alpha t)$. Since the latter expression tends to 0 as $t \rightarrow \infty$, it follows that the expectation on the right-hand side of (7) tends to 0 as $t \rightarrow \infty$. This fact, together with (8), gives

$$E[\exp(-\alpha T) \mathcal{I}(T < \infty)] = \exp(-\gamma_1 b),$$

which proves (1) for $b > 0$. The proof for $b < 0$, which uses $\gamma = \gamma_2$, is entirely analogous.

To complete the derivation of (1), we need to prove (4). Fix an integer $n \geq 1$ and set

$$X_j = B_{j/2^n} - B_{(j-1)/2^n}; \quad j = 1, 2, \dots$$

and

$$V_k = \exp(\gamma \sum_{j=1}^k X_j - \frac{1}{2} \gamma^2 k/2^n).$$

Since $E[\exp(\gamma X_j)] = \exp(\frac{1}{2} \gamma^2/2^n)$ and $Z_{k/2^n} = V_k$, it follows that

$$E[Z_{k/2^n}] = E[V_k] = 1 \text{ for all } k \geq 1. \quad (10)$$

Next, set

$$N = \sum_{l \geq 1} l \mathcal{I}((l-1)/2^n < T \leq l/2^n).$$

Observe that for every $k \geq 1$, the event $(N = k)$ depends only upon $X_1, \dots, X_k$. Given a positive integer m , set

$$N(m) = \min\{N, m\}. \quad (11)$$

The content of the following lemma is that the fixed k in (10) can be replaced by the random $N(m)$ in (11).

Lemma 1 $E(Z_{N(m)/2^n}) = E(V_{N(m)}) = 1$.

Proof. Since $1 \leq N(m) \leq m$, we have the identity

$$\sum_{k=1}^m \mathcal{I}(N(m) = k) = 1$$

and so, starting from (10) for $k = m$,

$$\begin{aligned} 1 &= E[V_m] \\ &= E[V_m \sum_{k=1}^m \mathcal{I}(N(m) = k)] \\ &= \sum_{k=1}^m E[V_m \mathcal{I}(N(m) = k)] \\ &= \sum_{k=1}^m E[(V_m/V_k) V_k \mathcal{I}(N(m) = k)] \\ &= \sum_{k=1}^m E[V_m/V_k] E[V_k \mathcal{I}(N(m) = k)] \\ &= \sum_{k=1}^m E[V_{m-k}] E[V_k \mathcal{I}(N(m) = k)] \\ &= \sum_{k=1}^m E[V_k \mathcal{I}(N(m) = k)] \\ &= E[V_{N(m)}]. \end{aligned}$$

The fifth equality in the preceding display follows because, for $k < m$,

$$V_m/V_k = \exp\left\{\lambda(X_{k+1} + \dots + X_m) - \frac{1}{2}\lambda^2(m-k)/2^n\right\}$$

depends upon $X_{k+1}, \dots, X_m$ only, while $V_k \mathcal{I}(N(m) = k)$ depends only upon $X_1, \dots, X_k$. The sixth equality (for $k < m$) follows because V_m/V_k has the same distribution as V_{m-k} while the seventh equality comes from (10). ■

Now set $m = \lfloor 2^n t \rfloor$ and let $n \rightarrow \infty$. Since $N/2^n \rightarrow T$, $m/2^n \rightarrow t$ and the map $s \mapsto Z_s$ is continuous, it follows that

$$Z_{N(m)/2^n} \rightarrow Z_{\tau(t)}. \quad (12)$$

Furthermore, for $b > 0$ and all n , the same argument that led to (9) gives

$$\begin{aligned} 0 &\leq Z_{N(m)/2^n} \leq \exp\{\gamma_1(B_{N(m)/2^n} - b) - \frac{1}{2}\gamma_1^2 N(m)/2^n\} \\ &\leq \exp(-\alpha m/2^n) \leq \exp(-\alpha t) < \infty. \end{aligned}$$

Thus, Lemma 1 and the bounded convergence theorem applied to (12) yields (4).

3. Density and distribution function

The Laplace transform of a real function $g(x)$, $0 < x < \infty$, is

$$\mathcal{L}(\lambda; g) = \int_0^\infty \exp(-\lambda x) g(x) dx.$$

The right-hand side of equation (1) can, for $\alpha > 0$, be expressed as $\mathcal{L}(\mu^2/2 + \alpha; g)$ where

$$g(x) = \exp(\mu b) |b| (2\pi x^3)^{-1/2} \exp(-b^2 x^{-1}/2), \quad x > 0,$$

see Abramowitz and Stegun (1970). Now,

$$\mathcal{L}(\mu^2/2 + \alpha; g) = \mathcal{L}(\alpha; f(\cdot; b, \mu)) \quad (13)$$

where

$$\begin{aligned} f(x; b, \mu) &= \exp(-\mu^2 x/2) g(x) \\ &= |b| (2\pi x^3)^{-1/2} \exp(-(|b|x^{-1/2} + \operatorname{sgn}(b\mu)|\mu|x^{1/2})^2/2), \end{aligned} \quad (14)$$

the last equality following after some algebraic manipulation and use of the fact that $b = \operatorname{sgn}(b) \times |b|$ for all real b . Since the right-hand side in (13) and the left-hand side in (1) represent the same object, it follows that the density of T is given by (14). To find the distribution function of T it suffices to verify

algebraically that the right-hand side of (14) equals

$$\frac{d}{dx} \left\{ \exp(2b\mu) \Phi(-|b|x^{-1/2} - \operatorname{sgn}(b\mu)|\mu|x^{1/2}) \right. \\ \left. + \Phi(-|b|x^{-1/2} + \operatorname{sgn}(b\mu)|\mu|x^{1/2}) \right\}$$

for $x > 0$.

4. Some consequences

The following known results are direct consequences of (1) and (2). Setting $\alpha = 0$ in (1) gives

$$P[T < \infty] = \exp((\operatorname{sgn}(\mu b) - 1)|\mu b|), \quad (15)$$

from which it is clear that $P[T < \infty] = 1$ if and only if μ and b have the same sign or if $\mu = 0$. Set

$$M = \sup_{s \geq 0} (B_s + \mu s).$$

Since the events $[T < \infty]$ and $[M \geq b]$ are identical when $b > 0$, it follows from (15) that M is exponentially distributed with mean $(2|\mu|)^{-1}$ when $\mu < 0$.

In the contrary case,

$$P[M = \infty] = 1.$$

Taking the derivative in (1) at $\alpha = 0$ gives

$$E[T] = \begin{cases} |b/\mu| & , \operatorname{sgn}(b\mu) = 1 \\ \infty & , \operatorname{sgn}(b\mu) \neq 1 \end{cases}$$

so that $E[T]$ is finite if and only if μ and b have the same sign. Clearly, the case $\mu = 0$ is special:

$$P[T < \infty] = 1 \text{ but } E[T] = \infty. \quad (16)$$

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